



Investigating the influence of erratic grid on stationary battery energy storage technologies in hybrid power systems: Techno-environ-economic perspectives

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ABSTRACT

Several obstacles impede renewable energy penetration into the global energy sector. Amongst the apparent challenges which require credible research attention, is the selection of appropriate energy storage technologies in the context of intermittent renewable resources and frequent grid outages. Given these, the present study investigated the technical, economic, and environmental effects of an erratic national grid on four distinct battery technologies in hybrid wind, solar, and diesel energy systems. The study was conducted at Baze University Abuja, Nigeria using the Hybrid Optimization Model for Electric Renewables software. The battery types considered are vanadium redox flow battery (VRB), lead-acid battery, nickel-iron battery, and lithium-ion battery (LIB). The VRB-based hybrid energy systems demonstrated superior performances in meeting the electricity demands of the university at the lowest net present cost, levelized cost of energy, and carbon dioxide emissions of \$6,328,003.00, \$0.0722/kWh, and 21,754 kg/year respectively. This battery technology is characterized by storage depletion, throughput, and losses of 796 kWh/year, 406,570 kWh/year, and 182,757 kWh/year respectively. The sensitivity analysis showed that the frequency and duration of main grid outages affect the optimal systems' economics, component sizes, battery energy losses, battery energy storage depletion, and renewable energy penetration.

1. Introduction

Electricity is the primary engine of socio-economic activities and its production is currently dominated globally by fossil resources, giving rise to the annual emissions of 35 Gt of carbon dioxides [1]. Furthermore, the electricity supply in Africa, especially the sub-Sahara region is enveloped with a "double tragedy". Apart from the low access rates due to insufficient generation capacity, locations having grid infrastructures face frequent interruptions whose duration sometimes lasts for hours or

days. The World Bank enterprise surveys have shown that, on average, more than 75 % of businesses or firms in sub-Saharan Africa are affected by power outages thereby forcing many of them to either rely on expensive diesel generators (DG), alter periods of operation, operate for few hours, or completely alter the production processes [2].

Renewable energy, particularly wind, and solar energy has widely been acknowledged as an efficient and effective remedy to the enormous challenges of greenhouse gas emissions, over-reliance on diminishing fossil resources, and increasing energy demand [3,4]. Given this, countries all over the world have continued to set ambitious goals to

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List of abbreviations			
LIB	Lithium-ion battery	DG	Diesel generator
LAB	Lead-acid battery	BG	Biogas generator
NIB	Nickel-iron battery	HOMER	Hybrid Optimization Model for Electric Renewables
VRB	Vanadium redox flow battery	DC	Direct current
ZBF	Zinc-bromine flow battery	AC	Alternating current
HRES	Hybrid renewable energy systems	SoC	State of Charge
LCOE	Levelised cost of energy	O&M	Operation and maintenance
NPC	Net present cost	RF	Renewable fraction
CRF	Capital recovery factor	MGOF	Mean grid outage frequency
WT	Wind turbine	LF	Load following
PV	Photovoltaic	CC	Cycle charging
		OS	Operating strategy
		GO	Generator order

intensify the penetration of renewable energy toward meeting national energy needs, specifically after the 2011 International Renewable Energy Agency (IRENA) inaugural congress whose principal theme revolved around the deployment of renewable energy and curtailment of climate change [5]. A typical example is China's new nearer-term target of attaining carbon neutrality before 2060 which includes the deployment of 1,200 GW of cumulative solar photovoltaic and wind capacity by 2030 [6]. There is a similar target by the Nigerian government termed "Vision 30:30:30" with the key objective of achieving 30 % of electricity generation capacity, equivalent to 13.8 GW from renewable energy by 2030 [7].

Renewable energy integration into the electricity grid still faces some significant challenges notwithstanding its benefits for the environment and sustainability. Thus, renewable energy sources provide an intermittent energy profile that causes power instability with the adverse effects of damaging the grid components such as motors and generators and a typical way to address this challenge is to introduce energy storage facilities into the system [8]. Mechanical energy storage technologies (pumped hydro, flywheel, and compressed air), electrical energy storage technologies (capacitors, super-capacitors, and superconducting magnetic energy storage), thermal energy storage technologies (high-temperature and low-temperature energy storage), and chemical energy storage technology (batteries, fuel cells, and hydrogen storage) are the predominant electricity storage technologies since the 19th century [9]. The commonly used energy storage systems in the broad literature on hybrid renewable power projects are pumped hydro, flywheel, batteries, and fuel cells [10–12].

The pumped hydro is capital-intensive and time-consuming to make it operational and also requires suitable topography which puts a significant geographical limitation on its deployment [9]. The flywheels have a long lifespan of about 20 years and an efficiency of about 93 % but have large standby losses, low energy density, and potentially dangerous failure modes [13]. Other drawbacks of flywheel energy storage systems, specifically when compared to battery and super capacitors are high cost, large noise, tedious maintenance effort, high self-discharge rate, and safety concerns [14]. Fuel cells (FC), on the other hand, have life cycles of at least 15 years, fast start-up features, and can be applied for small and large-scale energy storage and in simple modular use, but they are capital-intensive and suffer from a very low round trip efficiency [9]. Similar to batteries, supercapacitors provide an effective way to store extra energy, facilitate seamless power transfer, and reduce voltage fluctuations, particularly in applications that call for a quick cycle of charging and discharging. Meanwhile, supercapacitors' energy density is their greatest downside, implying a high initial cost of installation to achieve a similar density to that of batteries [15,16]. Typically, their rated voltage (less than 2.7 V) is quite low, hence, several series connections are required with adverse effects of increased internal resistance and reduced lifespans [16].

Battery energy storage system is an advanced technological solution

for stabilizing the grid and for providing backup power services in renewable energy systems. With obvious benefits like a swift response, controllability, and geographic independence, battery energy storage systems have garnered a lot of attention [17]. Apart from the benefits already mentioned, they also have a wide range of applications, including transmission upgrade deferral, reliability enhancement, long-term energy management, and power quality improvement [18].

Lead-acid batteries (LAB), which were invented in the early 20th century to supply energy to loads on a DC network, marked the beginning of the development of battery energy storage systems. Since then, different battery technologies have evolved and novel research in this context has taken center stage. Vanadium redox flow batteries, polysulfide-bromine flow batteries, sodium sulphur batteries, nickel-cadmium batteries, lithium-ion batteries, lithium metal polymer batteries, nickel-iron batteries, and zinc-bromine flow (ZBF) batteries are some of the other battery types that have found application for energy storage [19].

Rechargeable electrochemical devices for small and medium-scale energy storage applications have been dominated by LAB, which currently controls more than 60 % of the global battery market [20]. In case of cost-effective solutions and tolerance against mishandling, in which a limited life cycle and low energy density are not a concern, LAB is a technology of preference [21]. They, however, have a short lifespan, discharge explosive acid fumes, and gas, require regular maintenance, and exhibit poor performance in cold zones as a result, costly thermal energy management becomes a necessity [19,20]. Vanadium redox flow batteries, first developed in the 1980s, are rapidly penetrating the market as they currently constitute more than 20 MWh of installed storage capacity the world over [22]. They have proven to be robust in responding to fluctuating energy production from renewable resources and are used primarily for small to medium-scale storage applications. VRB, however, still faces some technical challenges such as poor electrolyte solubility and stability resulting in decreased energy densities, high capital and operating costs, and requirement for considerable layout due to their configuration [21,22]. Lithium-ion batteries (LIB) are another type of battery credited for the highest energy density, thus providing a huge prospect for deployment in a variety of energy storage, including stationary grid-scale energy storage applications [22]. They have low discharge rates, high levels of safety, superior charge retention, high cell voltages, and wide operating voltage, and when compared to aqueous technologies, they offer flexible packaging and are lightweight [21]. Meanwhile, lithium is extremely reactive on exposure to water or oxygen and must be packed with utmost care. More so, the costs and lifespan of this battery type are not as favorable compared to those of other battery technologies. In addition to the requirements for special packaging, there is a tendency for safety issues if the peak charge voltages are exceeded or the physical module is damaged leading to thermal runaway, discharging, explosion, and fire [21].

Since electricity generation from a single renewable energy source (i.

e. wind, hydro, biomass, solar, and tidal) cannot guarantee sustainability due to their stochastic nature and reliance on definite seasonal and weather circumstances, energy planners are more interested in amassing multiple energy sources with battery storage systems for complementary operation and cost-effective energy supply. Such systems (i.e. hybrid renewable energy systems) can be tied to the grid or operated in off-grid mode. For instance, Iqbal et al. [12] considered a fuel cell in an off-grid hybrid photovoltaic (PV), wind turbine (WT), and battery for maritime transportation in Pakistan. The system showed the prospects of producing 1,368,944 kWh/yr. Adetoro et al. [23] examined three varieties of dispatch strategies for optimal operation of grid-connected biogas generator (BG)/PV/WT/DG/battery systems. The study achieved a net present cost (NPC) of \$1.99 million and a levelized cost of energy (LCOE) of \$0.148/kWh. Roy et al. [11] employed a bilayered neural network and medium neural network to analyze the feasibility of deploying a decentralized BG/PV/WT/battery energy system to Henry Island in India. The proposed power system produces excess electricity of 2,838,968 kWh annually at an LCOE of \$0.4224 per kWh and an NPC of \$6.41 million. Kumar et al. [10] proposed an off-grid WT/PV/DG energy system that incorporates an electrolyzer and a hydrogen tank to reduce the demand for batteries. This system, which had an NPC of \$5.21 million and an LCOE of \$0.25/kWh, provides uninterrupted electricity by leveraging renewable resources. Akinte et al. [24] suggested the application of hybrid battery energy devices in HRES. The study utilized HOMER to investigate three control strategies consisting of the main grid, solar PV, BG, and hybrid energy storage system (iron flow, lithium, flywheel, and sodium sulphur) to power a thermal load.

In a similar vein, several studies have been conducted and published in the literature that evaluate the performance of various battery storage technologies in hybrid renewable energy systems (HRES). For instance, Iqbal et al. [12] used HOMER pro to optimize PV/WT, PV/WT/DG, and PV/WT/FC for a Shipboard in Pakistan. The analysis which considered a passenger vessel “FCS Alsterwasser” electricity consumption profile for the simulation, utilized two distinct battery technologies, namely LIB, and LAB. The LIB, LAB, and LIB respectively, presented the lowest LCOE of \$0.284/kWh for PV/WT, \$0.295/kWh for PV/WT/DG, and \$0.409/kWh for PV/Wind/FC. Alem et al. [17] analyzed LIB and LAB integrated with grid-tied PV systems for commercial loads in Ethiopia. The system with LIB was found to be more economical. The PV/ZBF system was more favored for prolonged grid interruption scenarios compared to PV/VBR, PV/LIB, PV/LAB, and PV/NIB in a HRES study undertaken by Barakat et al. [25] for El Dabaa, Matrouh Governorate, Egypt. Furthermore, in the absence of outages, the grid presented a lower LCOE of \$0.089/kWh and an NPC of \$121,382 for a long-term investment. An improved harmony search algorithm was utilized for the optimization of a HRES employing three different battery chemistries, namely, LAB, NIB, and LIB [26]. LAB was found most suitable to solve the power supply issue, save overall system costs, and improve system reliability. Mohammed et al. [27] used ant colony optimization, flower pollination optimization, particle swarm optimization, and genetic algorithms to evaluate the impacts of different battery capacities on NPC in off-grid WT/PV/DG energy systems. The results revealed an increasing pattern in NPC with increasing battery capacities. Dash et al. [28] proved the superiority of LIB over LAB and mechanical energy storage devices (flywheels and pumped hydro storage) in a hybrid tidal plant, WT, and biodiesel generator in India. The study performed a detailed analysis of the economic, energy, and environmental performances based on three distinct dispatch strategies - cycle charging (CC), load following (LF), and generator order (GO), respectively. The optimal configurations for the two scenarios (WT/biodiesel generator/tidal/LIB and WT/biodiesel generator/tidal/LAB), besides having the same NPC and LCOE of \$1.30 M and \$0.263 per kWh favored cycle charging strategy. Ramesh & Saini [29] performed a techno-economic analysis of employing LAB and LIB in off-grid PV/HT/hydro energy systems with and without scheduling DG. The LIB-based off-grid/PV/hydro system

offered the optimal solution. In Saini & Singal [30], four distinct battery modules (sodium-sulphur, LAB, LIB, and NIB) were considered for storage in an integrated HRES for twelve villages in the Uttarakhand State of India. Eight different architectures comprising a micro-hydropower plant, a PV array, and a BG were modeled, simulated, and optimized. The configuration incorporating sodium-sulphur batteries was found optimal with an LCOE of \$0.22/kWh and an NPC of \$0.82 million. Kumar & Saini [31] examined an optimal configuration of off-grid PV/WT/BG/DG energy systems in the context of different battery technologies to meet the energy demand of a location in India. The LF and CC strategies were considered in the analysis. Findings revealed that the NIB-based HRES with CC operating strategy is more efficient and also provided a minimum life cycle cost and an LCOE of \$633, 789.00 and \$0.23793 per kWh respectively. Foles et al. [32] investigated different scenarios of combining VRB and LIB in solar PV installation. A fifteen-year economic and energy analysis utilizing experimentally proven battery performance models to run the hybrid energy system was conducted. The PV-LIB proved superior in performance. Arévalo et al. [33] investigated the influence of LAB, VRB, LIB, and hydrogen-VRB on off-grid PV/hydro/DG energy systems based on indicators such as the LCOE, NPC, carbon dioxides emission, and unmet electric loads. The VRB-based system had a lower NPC and LCOE. A hybrid WT/PV/FC/electrolyzer/hydro storage energy system with multiple energy storage facilities, such as batteries and hydrogen storage systems was proposed by Yadav et al. [34]. Leveraging renewable resources at the project site, the optimal system was found as WT/PV/FC/electrolyzer/hydro storage. Babaei et al. [35] explored the techno-economic viability of deploying an off-grid HRES comprising WT/PV/DG energy system for a location in Canada. The analysis which considered four different batteries, presented an optimal system as having a 200 kW DG, 152 kW PV arrays, 190 kW bi-directional converter, and LIB of 853 kWh capacity.

1.1. Research gaps

Globally, the research on HRES has attracted a lot of attention in recent years with a large proportion not tilted toward grid-connected systems as revealed through the review of related studies. Furthermore, most of the studies that focused on grid-connected systems ignored the main grid failure scenarios. In emerging economies such as Nigeria, power outages, poor voltage profiles, load rationing, and system collapse incidences occur frequently. For instance, between 2014 and 2018, there were 14,945 and 5,581 outages on Nigeria's 132 kV and 330 kV transmission networks, respectively [36], while the national grid collapsed 695 times between 1987 and 2019 [5]. Finally, it is common in the previous studies on techno-economic analyses of HRES to adopt lead-acid batteries as a backup storage facility. Currently, there are varieties of battery technologies aggressively penetrating the market, and little to no attention has been focused on their inclusion in the energy storage options of HRES.

1.2. Contribution of the study

To address the aforementioned gaps, the current study examined the techno-economic and environmental impacts of utilizing distinct battery storage technologies in HRES (wind, solar, and diesel) coupled to an erratic national grid. The battery technologies considered in the analyses include lithium-ion batteries (LIB), nickel-iron batteries (NIB), lead-acid batteries (LAB), and vanadium redox flow batteries (VRB). Several design options were explored using a Hybrid Optimization Model for Electric Renewables (HOMER) software tool. The preference for HOMER is predicated on its utility, adaptability, and precision for the techno-economic and environmental evaluation of HRES [37]. Besides assessing the severity of erratic grids on these batteries using the mean grid outage frequency and mean repair time as indices, the effects of increasing electricity demands on the optimal systems of each of the

battery technologies were investigated through comprehensive sensitivity analyses. The technical analyses focused on the optimal capacities and sizes of the various components, energy production profiles, unmet loads, operating strategy, battery throughput, energy loss, storage depletion, and state of charge. The economic analyses covered the systems' net present cost (NPC), operating cost, replacement cost, levelized cost of energy (LCOE), salvage cost, and the cost of fuelling the diesel generator (DG). The environmental analyses expressed the extent of renewable energy penetration and the quantity of carbon emissions if the models investigated were to be deployed for the proposed project site. The present study, which can be adapted to a variety of circumstances, may help with choosing the HRES that is the most economical, reliable, and eco-friendly in complementing the operation of erratic national grids.

2. Materials and methods

2.1. System structure

The main components of the HRES are WT, solar PV panels, power converter, battery bank, main grid, and DG. The aggregate electricity demands of the study location are presented as the electrical load. The outputs of the DG, WT, and the grid, being in AC form are linked to the AC bus. Since the outputs of the solar panels and battery bank are in DC form, they are connected to the DC bus. The power converter ensures the flow of electricity through the AC and DC buses. It rectifies AC to DC to enable storage of energy in the battery and inverts DC to AC to facilitate the use of output energy from the battery and solar PV panel by the loads. Different battery technologies were introduced into the model since the study is aimed at investigating the most favorable battery technology in erratic grid scenarios. The architecture of the proposed model is shown in Fig. 1.

2.2. Project site

The grid-connected HRES explored in this study is proposed for Baze University Abuja, Nigeria. Established in 2011, the university which is located in the capital city of Nigeria currently operates six (6) Faculties and a School of Postgraduate Studies. It draws electricity from the grid through a 33 kV feeder originating from the Kukwaba 132/33 kV transmission substation in Games Village. The recurrent interruption of power supply from the grid has been a serious impediment to effective administrative activities, teaching, and cutting-edge research output. To tackle this challenge, the university recently awarded a contract to deploy 5 MW off-grid solar PV to power critical loads on its campus. Meanwhile, there is a widespread perception that where the main grid exists, off-grid HRES is not a cost-effective solution, and the main grid

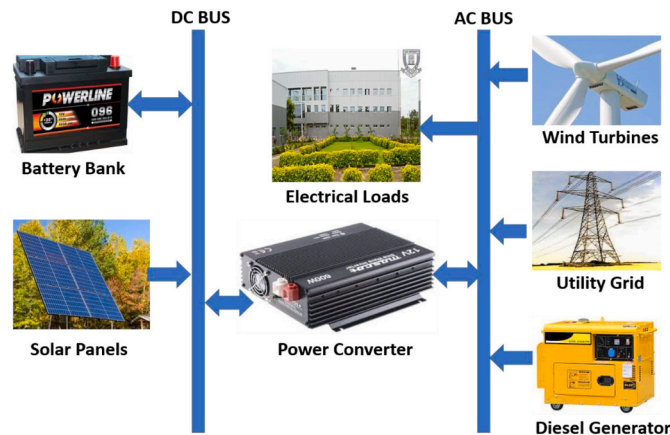


Fig. 1. Proposed hybrid power system.

tariff can be reduced when grid-connected HRES is employed [25]. Therefore, the present study will guide the decision-making of the university while exploring renewable energy options to solve the epileptic electricity supply currently being faced on the campus.

2.2.1. Meteorological profile of the project site

The renewable resources data of the university which include the solar irradiance, clearness index, wind speed, and temperature were assessed through the NASA database in the HOMER software tool [38]. These resources are represented in radar plots of Fig. 2. As seen in the Figures, wind speed at Baze University is relatively low. Meanwhile, the solar irradiance is high enough to guarantee sustainable electricity production using solar photovoltaic arrays. The highest potentials of solar irradiance, clearness index, and wind speed, respectively, are observed in March while the temperature is at its peak in February. The annual mean values of 24.71°C, 5.45 kWh/m²/day, and 3.01 m/s were considered for analyses.

2.2.2. Electricity demand of the university

The mean daily electricity demand of 3,804.12 kWh/day was assessed via the maximum demand energy meters in the university. The plots of this load demand considering the day-to-day random variability and time steps of 10 % and 20 % respectively are shown in Fig. 3. The graphs show that the periods of high electricity demand are when academic activities are ongoing and this corresponds to 9:00 a.m. to 6:00 p.m.

2.3. Modelling of the power system components

2.3.1. Wind energy system

Utilizing a wind turbine to convert the kinetic energy of the blowing wind into clean electricity is one way to harness wind energy. The blades rotate to trigger the rotor system, which converts the kinetic energy of the wind into electricity. The wind speed must be at least 3.0 m/s for a wind turbine system to provide the appropriate quantity of electric power [39]. As such, priority is often given to certain parameters which include the cut-out, rated, and cut-in speeds respectively, represented as V_{co} , V_r , and V_{in} while designing a wind turbine [40]. The mathematical modeling of the wind turbine's power output (P_{WT}) is given as [41]:

$$P_{WT}(t) = \begin{cases} 0 & \text{if } V < V_{in} \text{ or } V > V_{co} \\ V^3(t) \left(\frac{P_{WT(r)}}{V_r^3 - V_{in}^3} \right) - P_{WT(r)} \left(\frac{V_{in}^3}{V_r^3 - V_{in}^3} \right) & \text{if } V_{in} \leq V < V_e \\ P_{WT(r)} & \text{if } V_r \leq V \leq V_{co} \end{cases} \quad (1)$$

$P_{WT(r)}$ is the turbine-rated power. The wind speed (V) can be evaluated using eq. (2) [40,41].

$$V = v_{ref} \times \left(\frac{h_{hub}}{h_{ref}} \right)^\lambda \quad (2)$$

Typically, λ which is the ground surface roughness of the wind turbine ranges between 0.1 and 0.25. h_{hub} and h_{ref} are the hub and reference heights of the wind turbine respectively, while the v_{ref} represents the wind speed at the reference height. The cumulative energy produced by interconnected wind turbines can be computed using eq. (3) [27].

$$P_{WT}^T(t) = N_{WT} \times P_{WT}(t) \times \Delta t \quad (3)$$

Δt is the time step in an hour, N_{WT} specifies the number of installed turbines. The techno-economic data of the wind turbine model employed are presented in Table 1.

2.3.2. Diesel generator

It is impractical to produce a sufficient amount of energy using stochastic and volatile renewable resources to satisfy the occasionally varying electricity demands. More so, since the grid is erratic, there is a need for a backup diesel generator (DG) for enhanced system reliability.

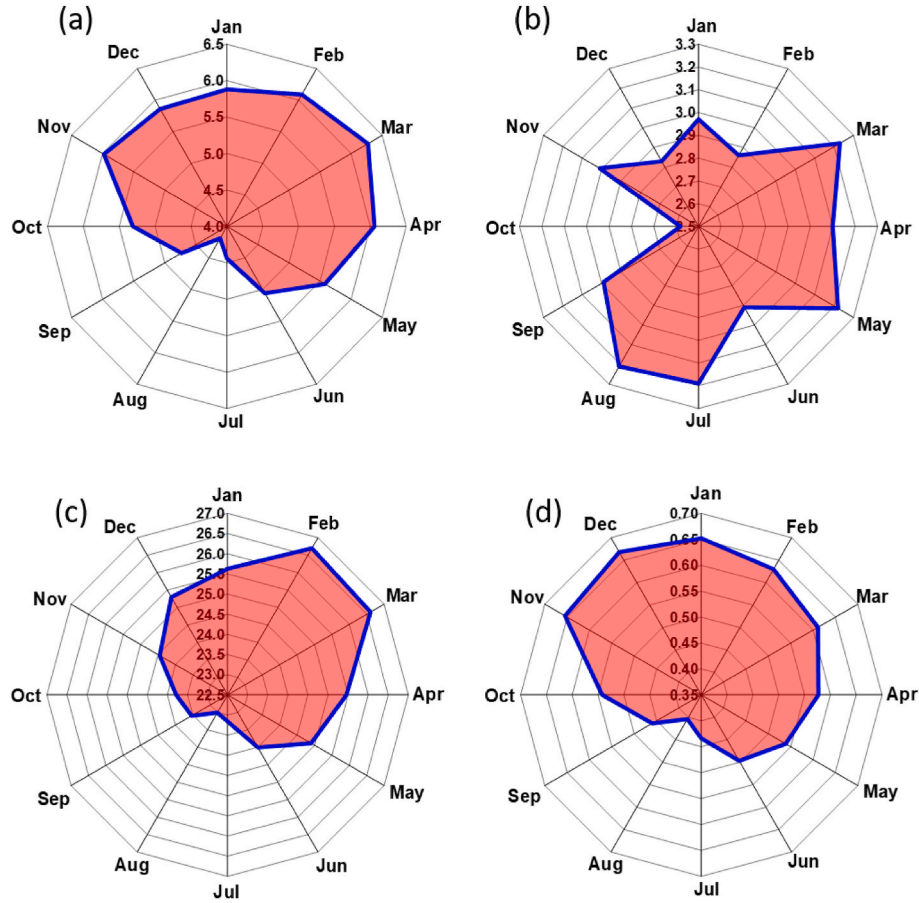


Fig. 2. Meteorological profiles of Baze University Abuja: (a) solar irradiance (b) wind speed (c) ambient temperature (d) clearness index.

The hourly fuel consumption of the DG, $DG_{FC}(t)$ in liter is expressed as [31]:

$$DG_{FC}(t) = (\alpha_{DG} \times P_{DG,gen}(t) + \beta_{DG} \times P_{DG,rat}(t)) \quad (4)$$

where α_{DG} (l/kWh) and β_{DG} (l/kWh) are the fuel consumption curve's coefficients usually specified by the DG manufacturer. The values for $\alpha_{DG} = 0.246$ l/kWh and $\beta_{DG} = 0.08145$ l/kWh have been chosen respectively [27].

The total annual fuel consumption is determined using eq. (5).

$$DG_{AFC} = \sum_{t=1}^{8760} DG_{FC}(t) \quad (5)$$

The techno-economic data of the DG model utilized in the analyses are presented in Table 2.

2.3.3. Solar energy system

The application of solar PV systems which generate electricity through the photovoltaic effects is the main emphasis of this study. The process through which sunlight, which is made up of energy particles called photons, strikes a solar panel and produces an electric current is known as the photovoltaic effect. A solar array is made up of several solar panels and produces far more energy than a single solar panel [44]. The total power generated based on the number of installed panels is determined using eq. (6) [27].

$$P_{PV}^T = N_{PV} \left[\frac{G_r}{G_R} P_M (1 + \tau(T_c - T_o)) \right] \quad (6)$$

where N_{PV} represents the number of panels installed, τ is the temperature correction at maximum power ($^{\circ}\text{C}$), T_c is the working temperature

of the panels, T_o is the reference temperature of the panels, solar radiation incident on the panel is represented by G_r , while the G_R represents the solar radiation at the point of reference given as 1000 W/m^2 . The techno-economic data of the solar PV modules used in the analyses are presented in Table 3.

2.3.4. Power converter

Converters are power electronic devices that facilitate the effective synchronization of AC to DC and vice versa. They function as rectifiers when converting AC signals to DC signals, and inverters when DC signals are converted to AC. Hence, to utilize the DC outputs from the solar PV and battery bank, an inverter is required while the energy outputs from the WT, DG, and grid supply require a rectifier to enable storage in DC form by the battery bank. The peak load demand ($P_L^{\max}$) determines the converter capacity. The power outputs during rectification and inversion respectively are determined using eqs. (7) and (8) [46].

$$P_{rec,out} = P_{AC} \times \eta_{rec} \quad (7)$$

$$P_{inv,out} = P_{DC} \times \eta_{inv} \quad (8)$$

where, η_{inv} , η_{rec} , P_{DC} , P_{AC} , $P_{inv,out}$, and $P_{rec,out}$ are the inverter efficiency (%), rectifier efficiency (%), rectifier power input (kW), inverter power input (kW), inverter power output (kW), and rectifier power output (kW) respectively. The techno-economic specifications of the power converter model employed are presented in Table 4.

2.3.5. Central grid

In Nigeria, the implementation of a new multi-year tariff order led to the present grid tariff plan termed "service reflective". There is no provision for net metering in this tariff structure and the unit cost of

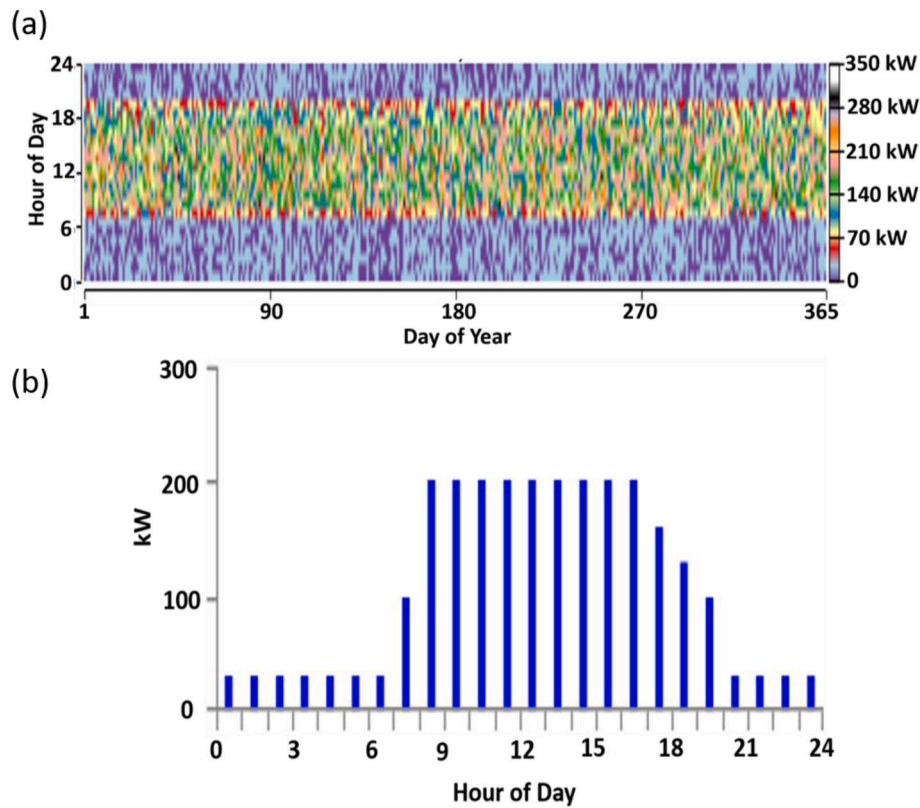


Fig. 3. Load profile of the university: (a) Seasonal loads (d) daily loads.

Table 1

Techno-economic data of wind turbine model [4,42].

Parameter	Value
Model	Bergy Excel 10-R
Manufacturer	Bergy Windpower
Rated capacity (kW)	10
Rated wind speed (m/s)	11.6
Start-up wind speed (m/s)	2.5
Cut-in wind speed (m/s)	2.2
Rotor diameter (m)	7
Lifetime (years)	20
Capital cost (\$/unit)	1980
Replacement cost (\$/year)	1980
O&M cost (\$/year)	25

Table 2

Techno-economic data of DG model [4,43].

Parameter	Value
Model	CAT-C9 Prime
Manufacturer	CAT
Rated capacity (kW)	275
Operational lifetime (Hours)	90,000
Capital cost (\$/kW)	250
Replacement cost (\$/kW)	160
O&M cost (\$/operating hour)	6

Table 3

Techno-economic data of solar PV modules [42,45].

Parameter	Value
Model	SunPower E20-327
Manufacturer	SunPower
Panel type	Flat plate
Rated capacity (kW)	30000
Nominal cell Operating temperature (°C)	45
Derating factor (%)	88
Efficiency at standard test condition (%)	20.4
Operational lifetime (years)	25
Capital cost (\$/kW)	925
Replacement cost (\$/kW)	800
O&M cost (\$/kW)	15

Table 4

Techno-economic data of power converter model [42,47].

Parameter	Value
Model	Leonics MTP-413F
Manufacturer	Leonics
Rated capacity (kW)	25
Inverter efficiency (%)	96
Rectifier efficiency (%)	95
Operational lifetime (years)	10
Capital cost (\$/kW)	700
Replacement cost (\$/kW)	500
O&M cost (\$/year)	10

electricity is independent of the seasons, off-peak, and peak periods [4,48]. Furthermore, the end-users are tied to various bands from “A-D” based on the hours of uninterrupted electricity they are scheduled to be supplied per day. The band “A” customers are scheduled for 20 h of steady power supply per day, and those on Bands “B” and “C” respectively, are meant to be supplied electricity for 16 h and 12 h per day. Bands “D” and “E” are at the bottom of the ladder as they are scheduled to enjoy a

maximum of 8 h of uninterrupted power supply daily.

Meanwhile, the grid regularly experiences failures that last for hours or even days, making the service-based structure ineffective. In this research, it is assumed that the grid outages occur twice daily, with an average of 18-h power availability, and 90 % repair time variability. The grid outages Dmap created by HOMER software from these data is shown in Fig. 4. The

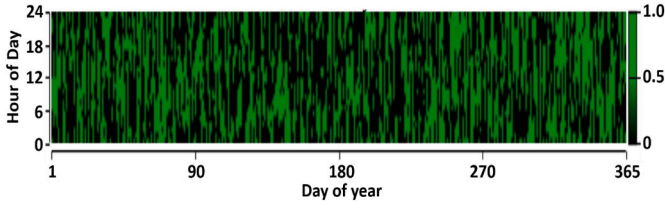


Fig. 4. Grid outage schedule.

black spots in the figure indicate when the grid is down and the normal operation of the grid is indicated by green color. The grid emission characteristics used for the simulation are according to Nigeria's electricity mix which includes CO₂ of 378 g/kWh; NO_x of 0.41 g/kWh; SO₂ of 1.9 g/kWh; CO of 0.03 g/kWh; and PM of 0.14 g/kWh [49]. More so, the present grid power tariff charged the customers under Band A in which the present study location falls is \$0.180 per kWh.

2.3.6. Battery energy storage system

An electrochemical storage system, known as the battery stores energy during a charging phase and releases it when the energy is required for utilization. The expression for the charging process of the battery bank is given as [31]:

$$E_{BS}(t) = (\sigma - 1) \times E_{BS}(t-1) + \eta_{BS} \times \left(E_G(t) - \frac{E_L(t)}{\eta_{conv}} \right) \quad (9)$$

where σ is the hourly self-discharge rate of the battery, $E_{BS}(t)$ and $E_{BS}(t-1)$ is the energy stored in the battery at the time (t) and time (t-1) respectively, $E_G(t)$ is the cumulative energy supplied the load by the hybrid power system, $E_L(t)$ represents the electrical energy demand of the study location, η_{conv} is the efficiency of the converter, and η_{BS} represents the roundtrip efficiency of the battery.

Furthermore, the electrical energy supplied by the hybrid power system is computed using eq. (10) [31].

$$E_G(t) = [E_{AC}(t) + E_{DC}(t)] \times \eta_{conv} \quad (10)$$

E_{AC} and E_{DC} are the energy produced in AC and DC forms respectively. The energy in AC form is the aggregate of the contribution from the WT, DG, and the grid, determined as:

$$E_{AC}(t) = E_{GRID}(t) + E_{WT}(t) + E_{DG}(t) \quad (11)$$

The energy in DC form is from the solar PV only given by:

$$E_{DC}(t) = E_{PV}(t) \quad (12)$$

in the event of grid failure, unfavorable meteorological conditions (solar irradiance and wind speed), and DG downtime, the battery bank discharges to provide the necessary energy deficit as expressed in eq. (13) [47].

$$E_{BS}(t) = (\sigma - 1) \times E_{BS}(t-1) - \eta_{BS} \times \left(E_G(t) - \frac{E_L(t)}{\eta_{conv}} \right) \quad (13)$$

Another crucial indicator for tracking a battery's condition and ensuring its safe functioning is its state of charge (SoC). It represents the proportion of the battery's capacity that is available for discharge to loads. Equation (14) is used to determine the SoC of the battery bank [47].

$$SOC(t) = SOC(t-1) \int_{t-1}^t \frac{P_b(t) \times \eta_{bat}}{V_{bus}} dt \quad (14)$$

η_{bat} stands for the battery's roundtrip efficiency (%), the bus voltage (V) is represented by V_{bus} , while t and $P_b(t)$ respectively represent the time and battery output power. The techno-economic specifications of the distinct battery technologies employed in the analyses are presented in Table 5.

Table 5

Specifications data of battery technologies used [25,40,41,50].

Parameter	Lead-Acid	Lithium-Ion	Nickel-Iron	Vanadium-Redox
Model	Kinetic	Idealized	Idealized	Idealized
Manufacturer	Hoppecke	Generic	Iron Edison	Gildemeister
Nominal Capacity (kWh)	7.15	100	12	100
Nominal Voltage (V)	2	600	12	48
Nominal capacity (Ah)	3,570	167	100	2,080
Peak charge current (A)	610	167	50	200
Peak discharge current (A)	610	500	50	313
Roundtrip efficiency (%)	86	90	80	64
Initial state of charge (%)	100	100	100	100
Minimum state of charge (%)	30	20	10	0
Capital cost (\$)	700	70,000	1,057	10,000
Replacement (\$)	645	1,000	1057	10,000
O&M cost (\$/year)	160	70,000	22	300
Operational lifetime (year)	20	15	30	20

2.4. System's economic parameters

2.4.1. Net present cost

Before the development and deployment of any power project, it is essential to analyze its economic potential throughout the operational lifetime. The net present cost (NPC) also known as the life cycle cost and the levelized cost of energy (LCOE) are the dominant economic criteria of merits for evaluating the performances of HRES. The aggregate of initial project costs, components replacement costs, and operation and maintenance (O&M) costs including the cost of fuelling the DG minus the present value of all profits earned during the lifetime of the project constitute an NPC. The mean cost of energy provided to the load by the HRES is known as the LCOE. Hence, it is key to minimize both the NPC and LCOE of HRES for techno-economic reasons and this is only possible through diligent usage of some economic criteria of interest. The total NPC of the HRES is computed using eq. (15) [17].

$$NPC = \frac{T_{ANC}}{CRF(i, n)} \quad (15)$$

where, T_{ANC} is the total annualized cost of the system's components in \$/year, $CRF(i, n)$ is the capital recovery factor, i and n represent the real interest rates in % and the project lifetime in year.

The annualized capital cost ($T_{ANC, cap}$), replacement cost ($T_{ANC, rep}$), and O&M cost ($T_{ANC, O\&M}$) of the various components in the system are all covered by T_{ANC} in eq. (16) [44].

$$T_{ANC} = \sum_n (T_{ANC, cap} + T_{ANC, O\&M} + T_{ANC, rep}) \quad (16)$$

2.4.2. Capital recovery factor

The capital recovery factor (CRF) is an economic index for the acquisition of capital as a uniform series of payments. It is a function of the investment cost and the period over which it must be paid off. The CRF is calculated as [33]:

$$CRF(i, n) = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (17)$$

The annual inflation rates (f) and the nominal interest rate ($inom$) define the real interest rate (i) as shown in eq. (18). The inflation rate and discount rate used in this study are 12 % and 6 % [44,51], respectively. The hybrid power system is proposed to have a lifespan of 25 years without any capacity shortage.

$$i = \frac{(i_{nom} - f)}{i + f} \quad (18)$$

2.4.3. Levelized cost of energy

The levelized cost of energy (LCOE) is the mean cost of energy delivered to the load by the HRES. The LCOE is determined using eq. (19) [17].

$$LCOE = \frac{T_{ANC}}{E_s} \quad (19)$$

E_s represents the total yearly energy served the load by the HRES.

2.4.4. Salvage cost

Salvage cost expresses the monetary value of a power system component that is still functional at the end of the lifetime of a project. HOMER utilizes eq. (20) to compute the worth of each component at the end of the project's life cycle [52].

$$S = C_{rep} \frac{R_{rem}}{R_{comp}} \quad (20)$$

Crep stands for the component replacement cost, Rrem represents the remaining lifetime of the component, and Rcom represents the component's lifetime.

2.5. Environmental emission model

Environmental emissions are also important factors when selecting an energy generation system. The combustion of fossil fuels for electricity generation releases various degrees of dangerous pollutants such as NO_x, particulate matter, carbon monoxides, carbon dioxides, sulphur dioxides, and unburned hydrocarbons. Therefore, preference is accorded to the energy systems having the least footprints of these pollutants for environmental sustainability. Renewable fraction (RF) is commonly employed as an index to provide a quick overview of the degree of renewable energy penetration in hybrid power systems operation. RF is a function of electricity delivered from renewable sources to the load. Its value must be high to curtail the impact of DG on the emission of greenhouse gases. The RF is expressed in eq. (21) [51].

$$RF(\%) = \left(1 - \frac{\sum E_{DG}(t)}{E_{WT}(t) + E_{PV}(t)} \right) \times 100 \quad (21)$$

where, $E_{DG}(t)$, $E_{WT}(t)$, and $E_{PV}(t)$ respectively, are the DG, WT, and solar PV output energy.

Furthermore, since DG is the major contributor to carbon dioxide emission in HRES, the hourly emission of this gas has a direct link with its fuel consumption on an hourly basis, estimated as [31]:

$$CO_2(t) = SE_{CO_2} \times F_{DG}(t) \quad (22)$$

where $FDG(t)$ is the hourly consumption of fuel by the DG, represents the specific CO₂ emissions per liter of fuel, and has a value of 2.7 kg/L.

The yearly CO₂ emissions are evaluated using eq. (23).

$$A_{CO} = \sum_{t=1}^{8760} CO_2(t) \quad (23)$$

2.6. Sensitivity analyses

The sensitivity variables used for the analyses are the mean grid outage frequency, mean repair time, and the load demand of the university. HOMER provides the flexibility to model HRES tied to unreliable grids. The outages are modeled as one or more steps in time during which no power can be wheeled to or from the grid. As such, the failure frequency per day, month, or year and the duration of the failure, respectively are defined by the mean grid outage frequency and mean repair time. It is projected that Baze University's need for electricity will

continue to rise since it is currently undergoing infrastructural expansion with more degree programmes springing up at both undergraduate and postgraduate levels. Besides determining the influence of the erratic grid on the different batteries, it is necessary to visualize how the increase in electricity demands will impact the investigated HRES. Therefore, the electricity demand was increased in a step of 10 % up to 60 % as a sensitivity variable. The sensitivity variables utilized for the analyses are presented in Table 6.

3. Results and discussion

The goal of this study is to investigate the techno-economic and environmental impacts of using four different battery technologies in HRES connected to an erratic grid supplying power to a university in Nigeria. The energy sources considered in the analyses are wind, diesel, solar, and main grid supply. HOMER software tool simulated both the feasible and unfeasible system architectures for the four distinct backup battery technologies, namely, LAB, LIB, NIB, and VRB with priority accorded to the system that meets the electricity demands at the lowest NPC, LCOE, and carbon footprints.

3.1. Effects of different batteries on components' sizes

The results of the most feasible systems generated via the HOMER software for each of the battery types, indicating the capacities of the power system components, LCOE, NPC, operating cost, and dispatch strategy, are shown in Table 7. LAB, LIB, and VRB-based HRES, respectively require PV panels of 1,000 kW while the NIB-based HRES requires PV panels of 5,000 kW to operate optimally. Similarly, the optimal configurations revealed that WT of 500 quantity is required for HRES utilizing LAB, LIB, and NIB while 150 WT is needed for operation if VRB-based HRES were to be employed to complement erratic grid operation. The operating strategy, sizes of the converter, and DG for the four battery types are load following (LF), 700 kW, and 275 kW respectively. In the LF operating scheme, whenever a generator operates, it produces sufficient electricity to satisfy the load demand while charging the battery is left to renewable power sources although the generator can still ramp up and evacuate power to the grid if its operation will offer some economic advantages. In case of insufficient or lack of electricity supply from the grid, DG, and renewable energy sources, 10 strings of LIB, 345 strings of LAB, 55 strings of VRB, and 10 strings of NIB respectively can be relied upon to satisfy the load imposed on the models. The large strings of LAB are attributed to its lower nominal voltage of 2 V compared to other battery types.

3.2. Effects of different batteries on system economics

The effects of different batteries on the optimal systems' economics are presented in Fig. 5. The dominant cost variables of the LAB-based HRES are the O&M costs of the LAB (\$3,051,005.87) and replacement costs of the power converter (\$1,659,717.22) and WT (\$1,473,802.29). The initial capital cost of the system is \$2,715,250.00 of which 36 %, 34 %, 18 %, 9 %, and 3 % respectively, will be expended on WT, PV panels, power converter, LAB, and DG. The DG and PV panels have no replacement costs implying their usefulness throughout the lifetime of the project. The cost of fueling the DG throughout the project's lifespan

Table 6
Sensitivity variables used for the analyses.

Sensitivity variables	Values
Mean grid outage frequency (1/yr)	365, 1095, 1460, 1825, 2190
Mean repair time (h)	1, 2, 4, 7, 8
Electricity demand (kWh/day)	4184.53, 4564.94, 4945.36, 5325.77, 5706.18, 6086.59

Table 7

Techno-economic results of the optimal systems.

Battery type	PV (kW)	WT (No.)	DG (kW)	Grid (kW)	CNV (kW)	Battery (strings)	NPC (M\$)	LCOE (\$/kWh)	OC (M\$/yr)	OS
LAB	1,000	500	275	999,999	700	345	9.74	0.0868	0.127	LF
VRB	1,000	150	275	999,999	700	55	6.33	0.0722	0.072	LF
LIB	1,000	500	275	999,999	700	10	9.34	0.0807	0.111	LF
NIB	5,000	500	275	999,999	700	10	23.90	0.1634	0.320	LF

PV – photovoltaic panel, WT – Wind turbine, DG – Diesel generator, CNV – Converter, NPC – Net present cost, LCOE – levelized cost of energy, OC – Operating cost, OS – operating strategy.

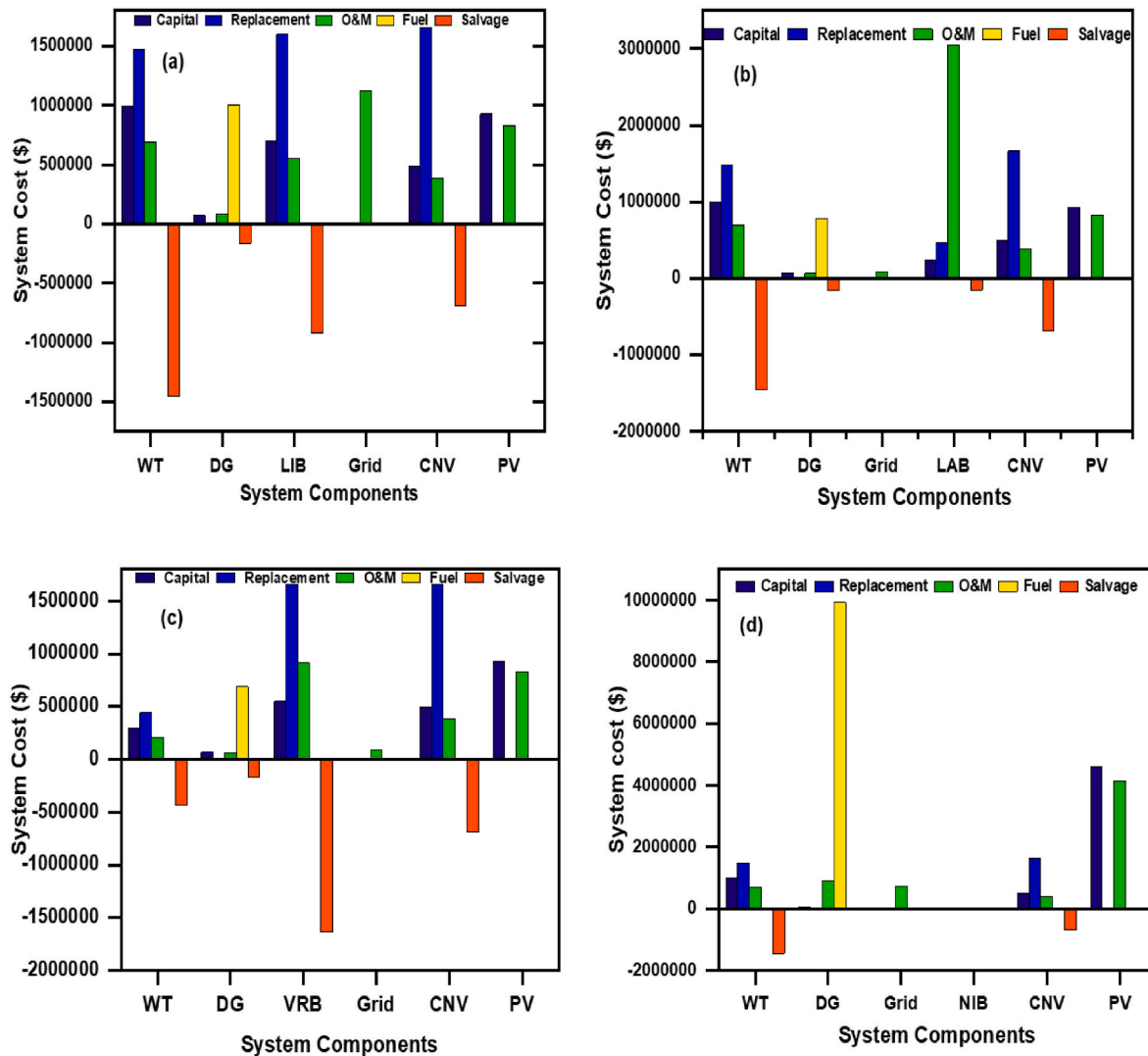


Fig. 5. Optimal systems' economics: (a) Grid/PV/WT/DG/LIB (b) Grid/PV/WT/DG/LAB (c) Grid/PV/WT/DG/VRB (d) Grid/PV/WT/DG/NIB.

is \$777,686.15 while the cost of O&M of the main grid is \$83,001.54. The salvage cost of the DG, battery, converter, and WT respectively are \$164,649.62, \$156,461.42, \$693,173.55, and \$1,455,664.46. Analysis of the overall costs of the system presented the battery storage as the most expensive component due to its high cost of O&M.

Similarly, the battery storage facility is the most cost-intensive component in LIB-based HRES. It has a capital cost of \$700,000.00, a replacement cost of \$1,598,749.42, and an O&M cost of \$552,718.46. The replacement costs of the converter, LIB, and wind turbines summed up to \$4,732,268.94 which is 51 % of the total NPC. Meanwhile, no cost will be incurred to replace the PV panels implying their prolonged usefulness over the project's lifetime. The salvage cost of this system is \$3,234,765.90 and is principally contributed to by the WT

(\$1,455,664.46), LIB (\$924,231.40), power converter (\$693,173.55), and DG (\$161,696.48). The cost of fueling the DG for this power system model is \$1,004,874.19.

In VRB-based HRES, the converter has the highest share of NPC of \$1,843,446.59 followed by PV panels (\$1,754,077.68), VRB (\$1,482,344.25), DG (\$646,693.69), WT (\$509,710.77) and the grid (\$91,729.81). Although the aggregate of the capital cost, replacement cost, and O&M costs of the VRB ranks topmost in the system economics, its high salvage cost (\$1,633,909.09) suppresses the NPC to pave the way to converter and PV panels to lead in the share of the total NPC. The replacement costs of the VRB, converter, and WT constitute 59 % of the total system costs less the salvage cost. The PV panels have no replacement cost and salvage cost while the DG will require

\$687,619.98 for fueling and its salvage cost is \$166,053.58.

The battery storage is relatively inexpensive in NIB-based HRES compared to LAB, LIB, and VRB. This battery model has a capital cost of \$10,570.00, an O&M cost of \$12,293.36, and a salvage cost of \$1,979.77. Furthermore, no replacement cost is associated with this battery signifying its extended service throughout the lifetime of the project. Furthermore, there is enormous pressure on DG to meet the electricity demand, and this results in the high fueling cost of \$9,918,574.15 which is about 42 % of the total NPC. Besides the DG fuel cost, the PV panels' capital cost (\$4,625,000.00) and O&M (\$4,145,388.42) are the most cost-intensive variables in the system economics. The lowest cost component in this power model is the O&M of the grid.

3.3. Electricity generation profiles

The quantum and periods of electricity generation from wind, solar, and diesel, and grid power purchases depend significantly on the battery technology employed as shown in Figs. 6 and 7. As seen in Fig. 6, in all the scenarios, the dominant energy source is solar followed by the wind while the DG and grid are rarely relied upon to supply the electricity demand. Besides the project site being located in a high solar radiation zone, the HOMER promotes penetration of solar PV systems because it is more efficient than wind energy systems. The LIB-HRES has no unmet loads and about 3.8 % of the cumulative electricity supply of 2,946,817 kWh/year is drawn from the main grid while DG has a share of just 0.8 %. The remainder of 56.7 % and 38.7 % of the total electricity generation, respectively is delivered by solar and wind energy systems. The highest electricity production from wind, solar, and DG respectively, are observed in May, December, and June as shown in Fig. 7. There is a capacity shortage of 1,189 kWh/year and an excess electricity production of 787,641 kWh/year associated with this power model. In a LAB-based HRES, a total of 2,837,129 kWh/year is produced with very

small fractions of 0.3 % and 0.7 % supplied by the main grid and DG respectively. There is no unmet load and electricity is generated in surplus (715,935 kWh/year) of consumption. The peak PV performance is observed in December (162,741.02 kW) while the wind energy system's peak performance is in May (120,574.12 kW). Furthermore, a substantial amount of electricity is available from the wind, and solar energy systems in January, March, April, and December with relative contribution from the grid that results in subduing the DG operation to zero power generation during the periods. A total of 2,038,820 kWh/year is produced by VRB-based HRES with contributions of 81.9 %, 16.8 %, 0.8 %, and 0.5 % respectively, made by the PV, WT, DG, and the grid. There is no power evacuation from the grid from January to May and in December. Similarly, the DG's output power is zero in November and December and from January to May. This can be attributed to the aggressive penetration of wind and solar energy systems during these periods. Excess electricity of 217,147 kWh/year, a capacity shortage of 1,327 kWh/year, and an unmet electric load of 67.2 kWh/year are associated with this system.

The exploitation of solar PV in NIB-based systems is massive (85.3 %) and this suppresses the wind energy system to a share of 11.6 % of the total electricity production of 9,796,085 kWh/year. Furthermore, this system has the highest DG penetration, accounting for its lower renewable fraction of 88.5 % compared to LAB, LIB, and VRB-based systems. Only 0.8 % (73,814.15 kW) of the total annual electricity is wheeled from the grid and the peak supply is observed in December similar to when PV output was at its zenith. In this power model, there is a surplus electricity generation of 7,082,897 kWh/year, a capacity shortage of 1,291 kWh/year, and an unmet load of 6.96 kWh/year.

3.4. Emission analyses

The main grid failure lasting over a long period, and the irregular and seasonal-dependence nature of renewable energy sources can affect a

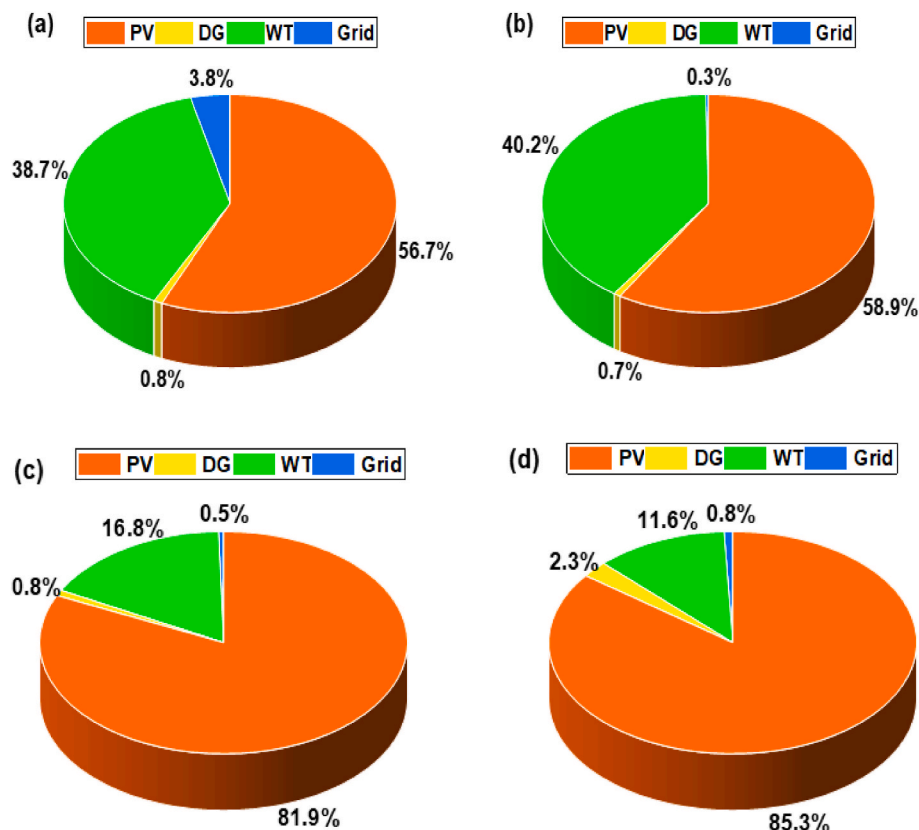


Fig. 6. Break down of electricity generation: (a) Grid/PV/WT/DG/LIB (b) Grid/PV/WT/DG/LAB (c) Grid/PV/WT/DG/VRB (d) Grid/PV/WT/DG/NIB.

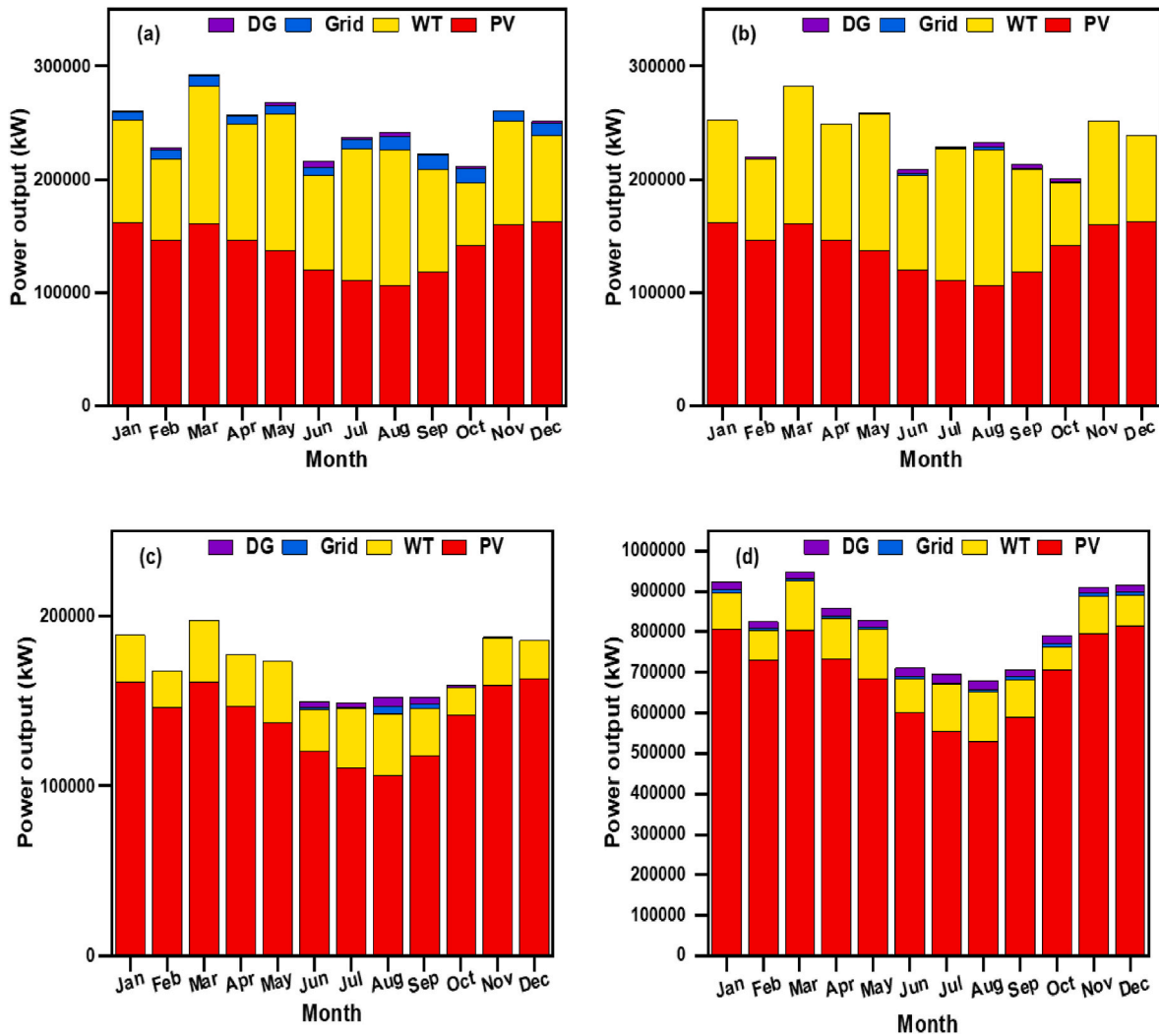


Fig. 7. Monthly electricity production profiles (a) Grid/PV/WT/DG/LAB (b) Grid/PV/WT/DG/LIB (c) Grid/PV/WT/DG/VRB (d) Grid/PV/WT/DG/NIB.

hybrid power system's performance especially if the batteries are continuously drained beyond their depth of discharge limits. Therefore, there is a need for dispatchable backup power from the DG to enhance the overall system's reliability. Diesel generators produce particulate matter, carbon monoxides, carbon dioxides, sulphur dioxides, nitrogen oxides, and unburned hydrocarbons that aggravate respiratory conditions. The release of these injurious air impurities and greenhouse gases correlates with the operational hour of the DG. For instance, NIB-based HRES's DG usage surpasses VRB, LAB, and LIB-based systems leading to more fuel consumption (i.e. 273 L/day) and consequent emission of these poisonous gases. The quantum of these gases associated with the NIB, LIB, LAB, and VRB-based systems is shown in Table 8. Since solar and wind are considered clean sources of energy, a measure of their penetration in relation to the penetration of fossil-based energy sources

in an HRES is expressed as RF. As seen in the Table, high penetration of renewable energy sources manifests when each of the LAB, VRB, and LIB is used as a backup storage facility compared to NIB. This is indicated by the RF of 98.7 %, 97.2 %, 98.3 %, 93.5 %, and 88.5 % respectively for LAB, VRB, LIB, and NIB-based HRES.

3.5. Battery performances

Table 9 presents the performance profiles of the distinct battery technologies in the operation of the HRES. The energy-in metric specifies the cumulative energy supplied per year for charging the battery bank. The peak energy-in is achieved for VRB while the NIB has the minimum energy-in. The annual charging energy for LAB and LIB, respectively are 59 % and 27.3 % of the annual charging energy for VRB.

Table 8
Emission analyses of the optimal systems.

Variable	LAB	NIB	VRB	LIB
Carbon Dioxide (kg/yr)	23,815	291,414	21,754	69,325
Nitrogen Oxides (kg/yr)	204	2,585	181	305
Carbon Monoxide (kg/yr)	12.9	163	11.4	19.7
Sulphur Dioxide (kg/yr)	66	780	61.9	279
Particulate Matter (kg/yr)	2.36	25.5	2.34	17.3
Unburned Hydrocarbons (kg/yr)	0.36	4.59	0.318	0.465
Renewable fraction (%)	98.7	88.5	98.3	93.5

Table 9
Performance profiles of different battery technologies in the operation of HRES.

Variable	VRB	LAB	LIB	NIB
Storage depletion (kWh/yr)	796	538	20.4	10.8
Expected life (yr)	20	13.7	15	26.2
Energy-in (kWh/yr)	507,217	273,831	138,387	5,049
Energy-out (kWh/yr)	325,256	235,994	124,568	4,049
Storage wear cost (\$/kWh)	0.0143	0.0687	0.246	0.0995
Annual throughput (kWh/yr)	406,570	254,479	131,306	4,527
Losses (kWh/yr)	182,757	38,375	13,840	1,011

The annual energy discharged by the battery banks defined as the energy-out follows a similar trend with the energy-in. The degree of battery operation also has an impact on its state of charge level (SoC). As a result, storage depletion has been taken into account, providing details regarding the variation in SoC between the start and end of the year. The VRB has the highest storage depletion followed by LAB, LIB, and then NIB. The battery bank experiences energy losses during charging and discharging due to storage inefficiency. The NIB has the minimum annual energy loss while the VRB has the maximum annual energy loss.

The cost of cycling energy through the battery bank is expressed by the storage wear cost. The LIB's storage wear cost is \$0.246 kWh, about 2.47, 3.58, and 17.2 times higher than that of NIB, LAB, and VRB respectively. The annual throughput of the battery storage technology defines the cumulative energy cycled through the battery bank in a year. The highest annual throughput is achieved for VRB (406570 kWh/year). The annual throughput for LIB, LAB, and NIB, respectively is 32.23 %, 62.6 %, and 1.1 % of the maximum annual throughput obtained for VRB. The lifespan of energy storage devices is a crucial performance index, and it has the most influence when evaluating economic efficiency. The aging and fading of the active materials resulting from the charge and

discharge cycles and the operational temperature are key contributors to the shortening of the lifespan of a battery. The LAB and NIB respectively have the minimum and maximum expected lifetime. Based on the results, NIB is the only battery technology that will not be replaced throughout the lifetime of the HRES project.

Another crucial indicator for tracking a battery's condition and ensuring its safe functioning is its state of charge (SoC). The SoC is a measure of the percentage of the battery capacity available for discharge to the loads. For a fully charged battery, the SoC is considered 100 % and for a fully discharged battery, it is 0 %. The Dmaps of the SoC of the different batteries in the operation of HRES are shown in Fig. 8. The Figures show that the SoC is not uniform throughout the year. For instance, on July 1st, the LIB attained 100 % SoC from 10:00 a.m.–9:00 p.m. and was able to operate between 20 % and 60 % of SoC from 12:00 a.m. –9:00 p.m. on July 5th. The NIB was at its peak during the period. However, it experienced a significant decay to 10 % around 5:00 a.m. on July 1st after which its SoC rose immediately to about 99.99 %. The LAB SoC ranges between 77 % and 100 % in this period. Meanwhile, the SoC toggles around 30 and 50 % throughout July 5th. The VRB has the lowest SoC compared to the other technologies. The maximum SoC achieved during this period is 50 % and values below 5 % are observed

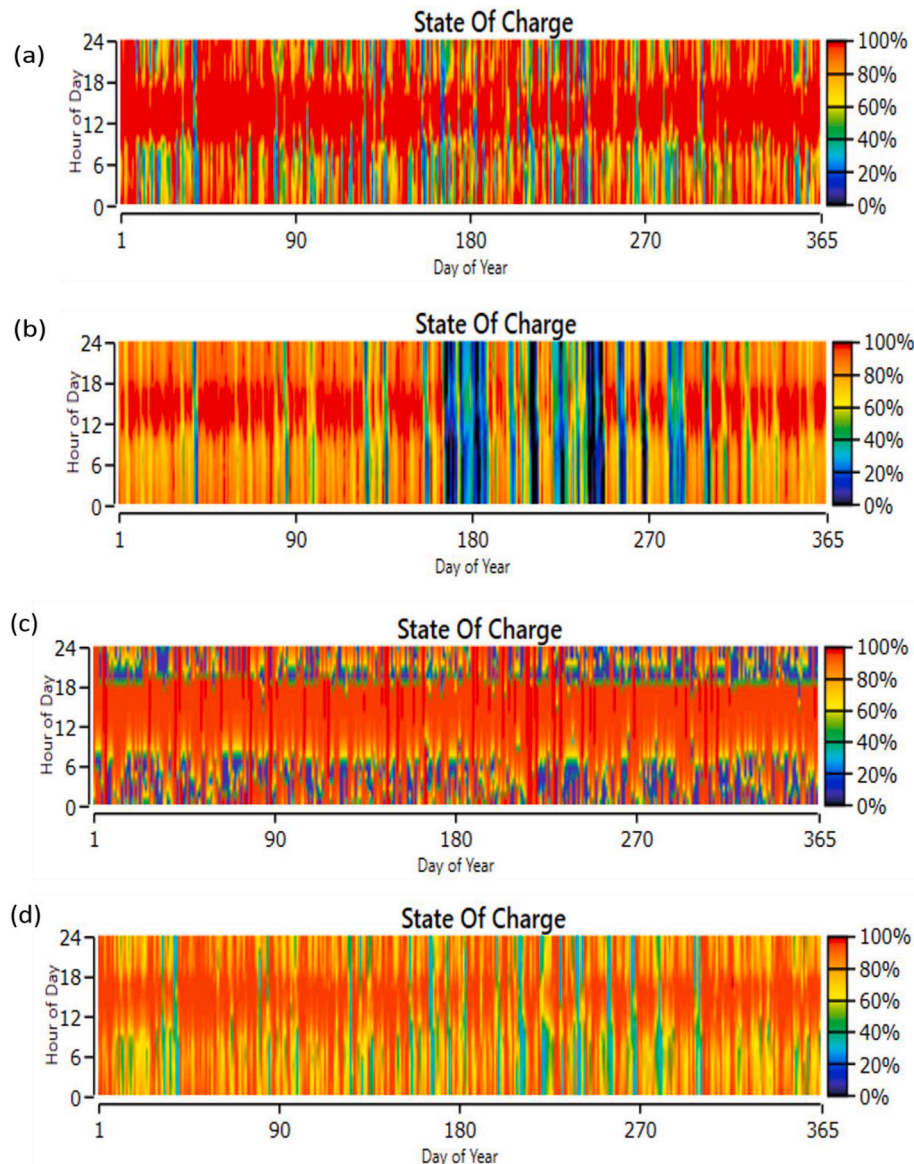


Fig. 8. State of charge of the batteries: (a) LIB, (b) VRB, (c) NIB, (d) LAB.

on July 5th.

3.6. Sensitivity analyses

The sensitivity analyses were conducted to visualize the effects of uncertainties that could occur during the lifetime of the proposed HRES project. Since the principal theme of the study is to assess the impacts of an unreliable grid on different battery technologies in HRES, the main grid failure frequency and the mean grid repair time were selected as sensitivity variables. Furthermore, an increase in load demand is anticipated owing to the massive infrastructural development the university is currently undergoing. As such, the load demand was varied in a step of 10 % up to 60 % to provide a bird-eye view of what the optimal system's performances will be under these scenarios. The results of the sensitivity analyses using the mean grid outage frequency, mean repair time and load demand are presented in [Tables A1–A3](#) in the supplementary document. As seen in the Tables, the variation of these parameters influences the overall technical, economic, and emission characteristics of the optimal HRES of the various battery technologies. Thus, the more these parameters are increased, the more the NPC, LCOE, and the capacities of the components.

As seen in [Table A1](#), with the mean grid outage frequency (MGOF) as the sensitivity variable, the number of WT and PV module capacities respectively rose to 2,000 and 10,000 kW in NIB-based HRES. These resulted in a significant surge of the systems' NPC and LCOE, thus providing the most costly HRES compared to other battery technologies.

Conversely, the VRB-based HRES has the lowest NPC and LCOE, but its battery losses are the highest compared to other battery types. The converter capacities are the same throughout except when the mean grid outage frequency value is 365 per year in LAB and NIB-based HRES. Similarly, it is shown in [Table A2](#) that, with the mean repair time as the sensitivity parameter, NIB-based HRES is the most cost-intensive system in terms of the NPC and LCOE while the VRB-based system ranks lowest. Under this condition, the LIB-based HRES utilizes more energy from the grid, yet has the lowest RF. Thus, the grid emission factors are more apparent in this battery technology. The VRB has more battery losses followed by the LAB, LIB, and NIB respectively.

The increase in electricity demands in a step of 10 % of up to 60 % led to an increase in the number of WT, capacities of the PV modules, renewable fraction, and overall costs of the system as shown in [Table A3](#) in the supplementary material. The battery losses, throughputs, storage depletions, storage wear costs, and projected lifespans of the batteries are sensitive to university electricity demand. The values of these variables relatively shoot up with an increasing university load. It is shown in [Fig. 9](#) (a) and (b) that, unlike LIB and VRB, the storage depletion of NIB and LAB are similar. The highest storage depletion in the range of 796–1098 kWh/year is observed with VRB while the lowest which is in the range of 15.8–63.90 kWh/year is associated with LIB. This indicates the robustness of LIB against deterioration from repeated charge/discharge, capacity degradation due to deep charge, and aging. Furthermore, as shown in [Fig. 9](#) (b), LIB and VRB respectively, have the lowest and highest annual storage throughputs compared to NIB and

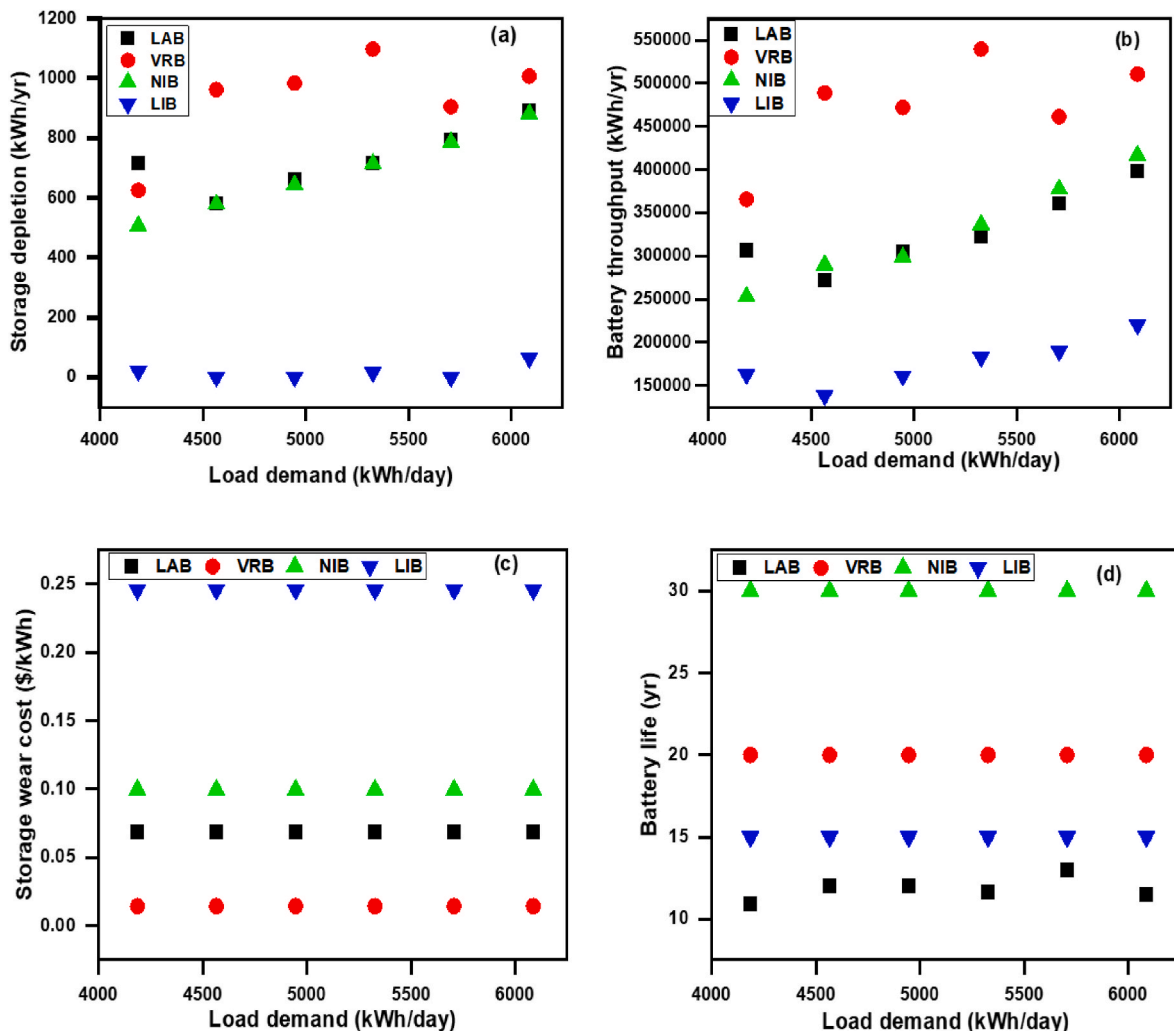


Fig. 9. Battery storage characteristics.

LAB in all the scenarios of university loads. The storage throughputs of LIB, NIB, LAB, and VRB respectively, when the campus load demand increases by 60 % (6087 kWh/day) is 220,274 kWh/year, 416,495 kWh/year, 397,405 kWh/year, and 510,542 kWh/year.

The annual storage wear cost (Fig. 9 (c)) is insensitive to the load demand. Thus, regardless of the university load, all the battery technologies maintained consistent storage wear costs. The storage wear cost in ascending order of magnitude is \$0.0143/kWh, \$0.0687/kWh, \$0.0995/kWh, and \$0.2460/kWh respectively, for VRB, LAB, NIB, and LIB. The term “expected life” describes how long a battery storage system is projected to operate according to its established characteristics and satisfy the necessary performance criteria. In Fig. 9(d), it is shown that unlike LAB whose lifetime varies with the load demand, the LIB, NIB, and VRB, each have a uniform lifespan regardless of the university electricity demand. The LIB, VRB, NIB, and LAB, respectively have a lifespan of 15 years, 20 years, 30 years, and 12 years. Since the proposed HRES is designed to last for 25 years, it is obvious that only NIB with a lifespan of 30 years will be functional throughout the project lifetime.

3.7. Comparison with other studies

The current study may not easily be compared to the relevant literature due to the complexity of system architectures, dimensions, load requirements, and sources of renewable energy available at various locations. However, several economic indicators may be utilized to compare the findings of this work with those of other studies in the literature. LCOE, which is computed by dividing the annualized cost of producing electricity by the total amount of power provided, can be a useful metric for assessing how cost-effective hybrid energy systems are. Table 10 compares some findings from the literature with the LCOE and winning battery technologies of the current study. The Table shows that the LCOE differs from one location to another. The LCOE, therefore, does not depend on the cumulative electricity demands but relies on the available resources at the project site, the components and fuel type selected, the components and fuel's economics, interest and inflation rates, and the constraints set by the hybrid power systems planner. The

Table 10
Optimal systems' performance comparison with similar studies.

Hybrid system configuration	Study location	Winning battery technology	LCOE (\$/kWh)	Reference
WT/DG	China	ZBF	0.471	Li et al. [41]
PV/BG	India	NIB	0.2178	Kumar & Saini [50]
HKT/PVDG	Ecuador	VRB	0.194	Arévalo et al. [33]
WT/PV/DG	Canada	LIB	0.321	Babaei et al. [35]
BG/WT/PV	India	NIB	0.23793	Kumar & Saini [31]
Grid/PV	Egypt	ZBF	0.089	Barakat et al. [25]
Grid/PV	Ethiopia	LIB	0.36	Alem et al. [17]
MHP/WT/PV/DG	India	LIB	0.106	Ramesh & Saini [29]
PV-WT-HT	India	LIB	0.158	Das et al. [53]
PV-battery	Portugal	LIB	0.30	Foles et al. [32]
WT/Bio-DG/Tidal	India	LIB and LAB	\$0.263 kWh	Dash et al. [28]
PV/WT	Pakistan	LIB	\$0.284/kWh	Iqbal et al. [12]
PV/WT/DG		LAB	\$0.295/kWh	
PV/Wind/FC		LIB	\$0.409/kWh	
Grid/WT/PV/DG	Nigeria	VRB	0.0722	Present study

winning battery technology in this study provides competitive or even better performance compared to other similar studies in the literature. Therefore, the results obtained in this study provide the University with sustainable energy options that are also cost-effective.

4. Conclusion

This study investigated the techno-economic and greenhouse gas emission effects of deploying different battery storage technologies in HRES (wind, solar, and diesel) connected to an unreliable national grid wheeling electricity to Baze University Abuja, Nigeria. The battery technologies considered in the analyses included lithium-ion batteries (LIB), nickel-iron batteries (NIB), lead-acid batteries (LAB), and vanadium redox flow batteries (VRB). The robust micropower optimization tool (HOMER) was employed to explore several design alternatives. Therefore, the following principal findings have been drawn from this study:

- (i) The VRB-based HRES provides the lowest NPC (\$6,328,003), LCOE (\$0.0722/kWh), and operating cost (\$72,319.88), while the NIB-based HRES has the highest NPC (\$23,895,120), LCOE (\$0.1634/kWh), and operating cost (\$320,430.80). The LIB and LAB-based HRES respectively ranked second and third in terms of cost-effectiveness.
- (ii) While the number of WT and capacities of the PV arrays differ, the capacities of converters (700 kW) and DG (275 kW) are the same regardless of the battery technologies employed in the HRES operation.
- (iii) High penetration of renewable energy sources is achieved when LAB, VRB, and LIB respectively, are employed as backup storage facilities compared to the NIB. This is demonstrated by the RF of 98.7 %, 97.2 %, 98.3 %, 93.5 %, and 88.5 % respectively, for LAB, VRB, LIB, and NIB-based HRES.
- (iv) The NIB-based HRES imposes enormous pressure on DG utilization. As a result, it accounts for the highest daily fuel consumption of 273 L and fuel cost of 9,918,574.15 during the project's lifetime. The fuel costs for VRB, LIB, and LAB respectively, are 6.93 %, 10.13 %, and 7.84 % of the fuel costs for NIB-based systems.
- (v) NIB-based HRES recorded the highest solar PV penetration (85.3 %) and this suppresses its wind energy system to a share of 11.6 % of the total annual electricity production of 9,796,085 kWh/day. It also presents the highest annual electricity production compared to other battery technologies.
- (vi) The HRES having the lowest and the highest carbon footprint respectively are the VRB and NIB-based systems. The annual emissions of carbon dioxides associated with the VRB-based system are 21,754 kg. The annual carbon dioxide emissions of NIB, LAB, and LIB-based systems respectively are approximately 7.46 %, 91.35 %, and 31.34 % more than the VRB-based system.
- (vii) The highest annual throughput is achieved for VRB (406570 kWh/year). The annual throughput for LIB, LAB, and NIB, respectively is 32.23 %, 62.6 %, and 1.1 % of the maximum annual throughput obtained for VRB. The energy loss in kWh/year and the storage depletion both follow a similar trend with the annual throughput.
- (viii) The increase in electricity demands, mean grid outage frequency, and the mean repair time in sensitivity analyses led to an increase in the number of WT and the capacities of the PV modules required to power the campus loads, as also the battery losses, renewable fraction, battery losses, and the overall costs of the system. Meanwhile, this has little to no effect on the capacities of the converter and DG set.

From the investigations, it can be concluded that the application of VRB as a backup energy storage facility in hybrid WT/PV/DG systems connected to an unreliable national grid is more cost-effective and

presents the minimum greenhouse gas emission compared to LIB, NIB, and LAB. Furthermore, the four scenarios of grid-connected HRES analyzed in this study have shown the prospects of lower tariffs compared to the utility grid tariff of \$0.180 charged by the Abuja Electricity Distribution Company Plc licensed to distribute electricity to customers in the nation's capital city, Abuja. The framework created through this research can be used as a guide by the energy audience, HRES planners, and decision-makers to decide how best to implement and integrate different battery technologies into existing power systems.

Further studies could examine the effects of other energy storage technologies, namely, pumped hydro storage, hydrogen storage, thermal energy storage, or the hybrid operation of a variety of battery technologies in HRES complementing unreliable grid operations.

CRediT authorship contribution statement

Usman Alhaji Dodo: Conceptualization, Formal analysis, Methodology, Resources, Software, Visualization, Writing – original draft, Writing – review & editing. **Babatunde Abiodun Salami:** Conceptualization, Writing – original draft, Writing – review & editing. **Faizah Mohammed Bashir:** Software, Validation, Writing – review & editing. **Haifa Youssef Hamdoun:** Software, Validation, Writing – review & editing. **Ibtihaj Saad Rashed Alsadun:** Software, Validation, Writing – review & editing. **Yakubu Aminu Dodo:** Software, Validation, Writing – review & editing. **A.G. Usman:** Software, Validation, Writing – review & editing. **Sani I. Abba:** Conceptualization, Formal analysis, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

We have no conflict of interest associated with this publication, and the publication of this article is approved by all authors and that, if accepted, it will not be published elsewhere in the same form, in English or any other language.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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References

- [1] Panda A, Dauda AK, Chua H, Tan RR, Aviso K. Advances, Recent in the integration of renewable energy sources and storage facilities with hybrid power systems. *Clean Eng Technol* 2023;1–55. <https://doi.org/10.1016/j.energy.2022.125507>.
- [2] Nduhuura P, Garschagen M, Zerga A. Impacts of electricity outages in urban households in developing countries: a case of Accra, Ghana. *Energies* 2021;14:1–26.
- [3] Huang WC, Zhang Q, You F. Impacts of battery energy storage technologies and renewable integration on the energy transition in the New York State. *Adv Appl Energy* 2023;9:1–20. <https://doi.org/10.1016/j.adapen.2023.100126>.
- [4] Dodo UA, Ashigwuike EC. Techno-economic and environmental analysis of utility-scale hybrid renewable energy system integrating waste-to-energy plant to complement an unreliable grid operation. *Energy, Ecol Environ* 2023;1–18. <https://doi.org/10.1007/s40974-023-00276-7>.
- [5] Dodo UA, Ashigwuike EC, Eronu EM. Renewable energy readiness in Nigeria: a review focusing on power generation. *Uniaibuja J Eng Technol* 2021;1:115–44.
- [6] International Energy Agency. *Renewables: analysis and forecast to 2026*. 2021.
- [7] Bugaje A-AB, Dioha MO, Saka B, Isah A, Anene-Nzeli O. Nigeria's electricity vision 30-30-30: exploring the techno-economic and environmental implications. *ECOWAS Sustain Energy J* 2022;3.
- [8] Zhao C, Andersen PB, Træholt C, Hashemi S. Grid-connected battery energy storage system: a review on application and integration. *Renew Sustain Energy Rev* 2023;182:1–19. <https://doi.org/10.1016/j.rser.2023.113400>.
- [9] Amir M, Deshmukh RG, Khalid HM, Said Z, Raza A, Muyeen SM, et al. Energy storage technologies: an integrated survey of developments, global economical/environmental effects, optimal scheduling model, and sustainable adaption policies. *J Energy Storage* 2023;72:1–34. <https://doi.org/10.1016/j.est.2023.108694>.
- [10] Kumar PH, Gopi RR, Rajarajan R, Vaishali NB, Vasavi K, Kumar PS. Prefeasibility techno-economic analysis of hybrid renewable energy system. *E-Prime - Adv Electr Eng Electron Energy* 2024;7:1–19. <https://doi.org/10.1016/j.prime.2024.100443>.
- [11] Roy D, Zhu S, Wang R, Mondal P, Ling-Chin J, Roskilly AP. Techno-economic and environmental analyses of hybrid renewable energy systems for a remote location employing machine learning models. *Appl Energy* 2024;361:1–17. <https://doi.org/10.1016/j.apenergy.2024.122884>.
- [12] Iqbal R, Liu Y, Zeng Y, Zhang Q, Zeeshan M. Comparative study based on techno-economics analysis of different shipboard microgrid systems comprising PV/wind/fuel cell/battery/diesel generator with two battery technologies: a step toward green maritime transportation. *Renew Energy* 2024;221:1–17. <https://doi.org/10.1016/j.renene.2023.119670>.
- [13] Rahman MM, Oni AO, Gemechu E, Kumar A. Assessment of energy storage technologies: a review. *Energy Convers Manag* 2020;223:1–28. <https://doi.org/10.1016/j.enconman.2020.113295>.
- [14] Koohi-Fayegh S, Rosen MA. A review of energy storage types, applications and recent developments. *J Energy Storage* 2020;27:1–23. <https://doi.org/10.1016/j.est.2019.101047>.
- [15] Mbendane B, Abe B, Richards C. Analysis of supercapacitors in renewable energy systems for managing power fluctuations. *Int J Energy Res* 2024;24:1–13. <https://doi.org/10.1155/2024/4575761>.
- [16] Zhang J, Gu M, Chen X. Supercapacitors for renewable energy applications: a review. *Micro Nano Eng* 2023;21:1–20. <https://doi.org/10.1016/j.mne.2023.100229>.
- [17] Alem AK, Coosemans T, Messagie M, Jemal T, Ayele H, Mierlo J Van, et al. Techno-economic analysis of lithium-ion and lead-acid batteries in stationary energy storage application. *J Energy Storage* 2021;40:1–19. <https://doi.org/10.1016/j.est.2021.102748>.
- [18] Das CK, Bass O, Kothapalli G, Mahmoud TS, Habibi D. Overview of energy storage systems in distribution networks: placement, sizing, operation, and power quality. *Renew Sustain Energy Rev* 2018;91:1205–30. <https://doi.org/10.1016/j.rser.2018.03.068>.
- [19] Kim J, Suharto Y, Daim TU. Evaluation of electrical energy storage (EES) technologies for renewable energy: a case from the US Pacific Northwest. *J Energy Storage* 2017;11:25–54. <https://doi.org/10.1016/j.est.2017.01.003>.
- [20] Abidin Z, Khalilpour KR. Single and polystorage technologies for renewable-based hybrid energy systems. Elsevier Inc.; 2019. <https://doi.org/10.1016/B978-0-12-813306-4.00004-5>.
- [21] Mexis I, Todeschini G. Battery energy storage systems in the United Kingdom: a review of current state-of-the-art and future applications. *Energies* 2020;13:1–30. <https://doi.org/10.3390/en13143616>.
- [22] Johnson SC, Davidson FT, Rhodes JD, Coleman JL, Bragg-Sittion SM, Dufek EJ, et al. Selecting favorable energy storage technologies for nuclear power. Elsevier Inc.; 2019. <https://doi.org/10.1016/B978-0-12-813975-2.00005-3>.
- [23] Adetoro SA, Olatomiwa L, Tsado J, Dauda SM. Techno-economic and environmental impact assessment of a hybrid renewable energy system employing an enhanced combined dispatch strategy. *Green Energy Resour* 2023;1:1–15. <https://doi.org/10.1016/j.gerr.2023.100044>.
- [24] Akinte OO, Plangklang B, Prasartkaew B, Aina TS. Energy storage management of a solar photovoltaic-biomass hybrid power system. *Energies* 2023;16:1–32. <https://doi.org/10.3390/en16135122>.
- [25] Barakat S, Emam A, Samy MM. Investigating grid-connected green power systems' energy storage solutions in the event of frequent blackouts. *Energy Rep* 2022;8:5177–91. <https://doi.org/10.1016/j.egy.2022.03.201>.
- [26] Sun X, He H, Ma L. Harmony search meta-heuristic algorithm based on the optimal sizing of wind-battery hybrid micro-grid power system with different battery technologies. *J Energy Storage* 2024;75:1–14. <https://doi.org/10.1016/j.est.2023.109582>.
- [27] Mohammed YS, Adetokun BB, Oghoroda O, Oshiga O. Techno-economic optimization of standalone hybrid power systems in context of intelligent computational multi-objective algorithms. *Energy Rep* 2022;8:11661–74. <https://doi.org/10.1016/j.egy.2022.09.010>.
- [28] Dash RL, Mohanty B, Hota PK. Energy, economic and environmental (3E) evaluation of a hybrid wind/biodiesel generator/tidal energy system using different energy storage devices for sustainable power supply to an Indian archipelago. *Renew Energy Focus* 2023;44:357–72. <https://doi.org/10.1016/j.ref.2023.01.004>.
- [29] Ramesh M, Saini RP. Effect of different batteries and diesel generator on the performance of a stand-alone hybrid renewable energy system. *Energy Sources, Part A Recover Util Environ Eff* 2020;1–23. <https://doi.org/10.1080/15567036.2020.1763520>.
- [30] Saini V, Singal SK. Optimal sizing of an integrated renewable energy system and effective utilization of surplus energy in electric vehicle charging. *Eng Sci Technol an Int J* 2023;48:1–21. <https://doi.org/10.1016/j.jestch.2023.101564>.
- [31] Kumar PP, Saini RP. Optimization of an off-grid integrated hybrid renewable energy system with various energy storage technologies using different dispatch strategies. *Energy Sources, Part A Recover Util Environ Eff* 2020;1–29. <https://doi.org/10.1080/15567036.2020.1824035>.

- [32] Foles A, Fialho L, Horta P, Collares-Pereira M. Economic and energetic assessment of a hybrid vanadium redox flow and lithium-ion batteries, considering power sharing strategies impact. *J Energy Storage* 2023;71:1–12. <https://doi.org/10.1016/j.est.2023.108167>.
- [33] Arévalo P, Darío B, Lata-García J, Jurado F. Energy control and size optimization of a hybrid system (Photovoltaic-Hidrokinetic) using various storage technologies. *Sustain Cities Soc* 2020;52:1–15. <https://doi.org/10.1016/j.scs.2019.101773>.
- [34] Yadav S, Kumar P, Kumar A. Techno-economic assessment of hybrid renewable energy system with multi-energy storage system using HOMER. *Energy* 2024;297:1–21. <https://doi.org/10.1016/j.energy.2024.131231>.
- [35] Babaei R, Ting DS, Cariveau R. Feasibility and optimal sizing analysis of stand-alone hybrid energy systems coupled with various battery technologies: a case study of Pelee Island. *Energy Rep* 2022;8:4747–62. <https://doi.org/10.1016/j.egyr.2022.03.133>.
- [36] Adelakun NO, Olanipekun BA. Outage analysis and system disturbances on 330 kV and 132 kV transmission system in Nigeria. *Eur J Eng Res Sci* 2020;5:100–2. <https://doi.org/10.24018/ejers.2020.5.1.1722>.
- [37] Malik P, Awasthi M, Sinha S. A techno-economic investigation of grid integrated hybrid renewable energy systems. *Sustain Energy Technol Assessments* 2022;51:1–15. <https://doi.org/10.1016/j.seta.2022.101976>.
- [38] Dodo UA, Ashigwuie EC, Emechebe JN. Optimization of standalone hybrid power system incorporating waste-to-electricity plant: a case study in Nigeria. 2022 IEEE Niger. 4th Int. Conf. Disruptive Technol. Sustain. Dev. Lagos, Nigeria: IEEE; 2022. p. 1–5. <https://doi.org/10.1109/NIGERCON54645.2022.9803081>.
- [39] Balat M. A review of modern wind turbine technology. *Energy Sources, Part A Recover Util Environ Eff* 2009;31:1561–72. <https://doi.org/10.1080/15567030802094045>.
- [40] Muna YB, Kuo CC. Feasibility and techno-economic analysis of electric vehicle charging of PV/Wind/Diesel/Battery hybrid energy system with different battery technology. *Energies* 2022;15:1–24. <https://doi.org/10.3390/en15124364>.
- [41] Li C, Zhou D, Wang H, Lu Y, Li D. Techno-economic performance study of stand-alone wind/diesel/battery hybrid system with different battery technologies in the cold region of China. *Energy* 2020;192:1–24. <https://doi.org/10.1016/j.energy.2019.116702>.
- [42] Khan FA, Pal N, Saeed SH, Yadav A. Modelling and techno-economic analysis of standalone SPV/Wind hybrid renewable energy system with lead-acid battery technology for rural applications. *J Energy Storage* 2022;55:1–23. <https://doi.org/10.1016/j.est.2022.105742>.
- [43] Chauhan A, Khan MT, Srivastava A, Tripathi A, Hussain SMS, Ustun TS. Techno-economic assessment and environmental analysis of an optimal hybrid system under novel demand response scheme for a remote region of India. *Energy Rep* 2022;8:284–91. <https://doi.org/10.1016/j.egyr.2022.05.200>.
- [44] Dodo UA, Ashigwuie EC, Emechebe JN. Techno-economic evaluation of municipal solid waste-fueled biogas generator as a backup in a decentralized hybrid power system. *Process Integr Optim Sustain* 2022;1–16. <https://doi.org/10.1007/s41660-022-00223-9>.
- [45] Bagheri M, Delbari SH, Pakzadmanesh M, Kennedy CA. City-integrated renewable energy design for low-carbon and climate-resilient communities. *Appl Energy* 2019;239:1212–25. <https://doi.org/10.1016/j.apenergy.2019.02.031>.
- [46] Khosravani A, Safaei E, Reynolds M, Kelly KE, Powell KM. Challenges of reaching high renewable fractions in hybrid renewable energy systems. *Energy Rep* 2023;9:1000–17.
- [47] Cano A, Arevalo P, Jurado F. Energy analysis and techno-economic assessment of a hybrid PV/HKT/BAT system using biomass gasifier: Cuenca-Ecuador case study. *Energy* 2020;202:1–16. <https://doi.org/10.1016/j.energy.2020.117727>.
- [48] Sanni SO, Oricha JY, Oyewole TO, Bawonda FI. Analysis of backup power supply for unreliable grid using hybrid solar PV/diesel/biogas system. *Energy* 2021;227:1–15. <https://doi.org/10.1016/j.energy.2021.120506>.
- [49] Jumare IA, Bhandari R, Zerga A. Assessment of a decentralized grid-connected photovoltaic (PV)/wind/biogas hybrid power system in northern. *Energy Sustain Soc* 2020;10:1–25.
- [50] Kumar PP, Saini RP. Optimization of an off-grid integrated hybrid renewable energy system with different battery technologies for rural electrification in India. *J Energy Storage* 2020;32:1–23. <https://doi.org/10.1016/j.est.2020.101912>.
- [51] Salisu S, Mustafa MW, Olatomiwa L, Mohammed OO. Assessment of technical and economic feasibility for a hybrid PV-wind-diesel-battery energy system in a remote community of north-central Nigeria. *Alex Eng J* 2019;58:1103–18. <https://doi.org/10.1016/j.aej.2019.09.013>.
- [52] Riayatsyah TMI, Geumpana TA, Rizwanul Fattah IM, Rizal S, Indra Mahlia TM. Techno-economic analysis and optimisation of campus grid-connected hybrid renewable energy system using HOMER grid. *Sustain Times* 2022;14:1–18. <https://doi.org/10.3390/su14137735>.
- [53] Das S, Pradhan S, De S. Multi-criteria decision making for the most suitable combination of energy resources for a decentralized hybrid energy solution with green hydrogen as the storage option. *Energy Convers Manag* 2023;285:1–17. <https://doi.org/10.1016/j.enconman.2023.117028>.